



CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 2

**CANDIDATE
NAME**

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CLASS

2T

**INDEX
NUMBER**

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BIOLOGY

Paper 2 STRUCTURED QUESTIONS

9744/02

21ST AUGUST 2017

2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid. **DO NOT WRITE IN ANY BARCODES.**

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [12]	
2 [12]	
3 [13]	
4 [12]	
5 [9]	
6 [13]	
7 [12]	
8 [7]	
9 [10]	
TOTAL P2 [30%]	100

This document consists of **19** printed pages and **1** blank page.

[Turn over

Answer **all** questions.

- 1 Fig. 1.1 is a schematic diagram showing the transport pathways of extracellular and intracellular materials for digestion in a mammalian cell. Depending on the types of digested material, three possible pathways are initiated to deliver these materials for digestion within lysosomes, of which one is labelled as **A**.

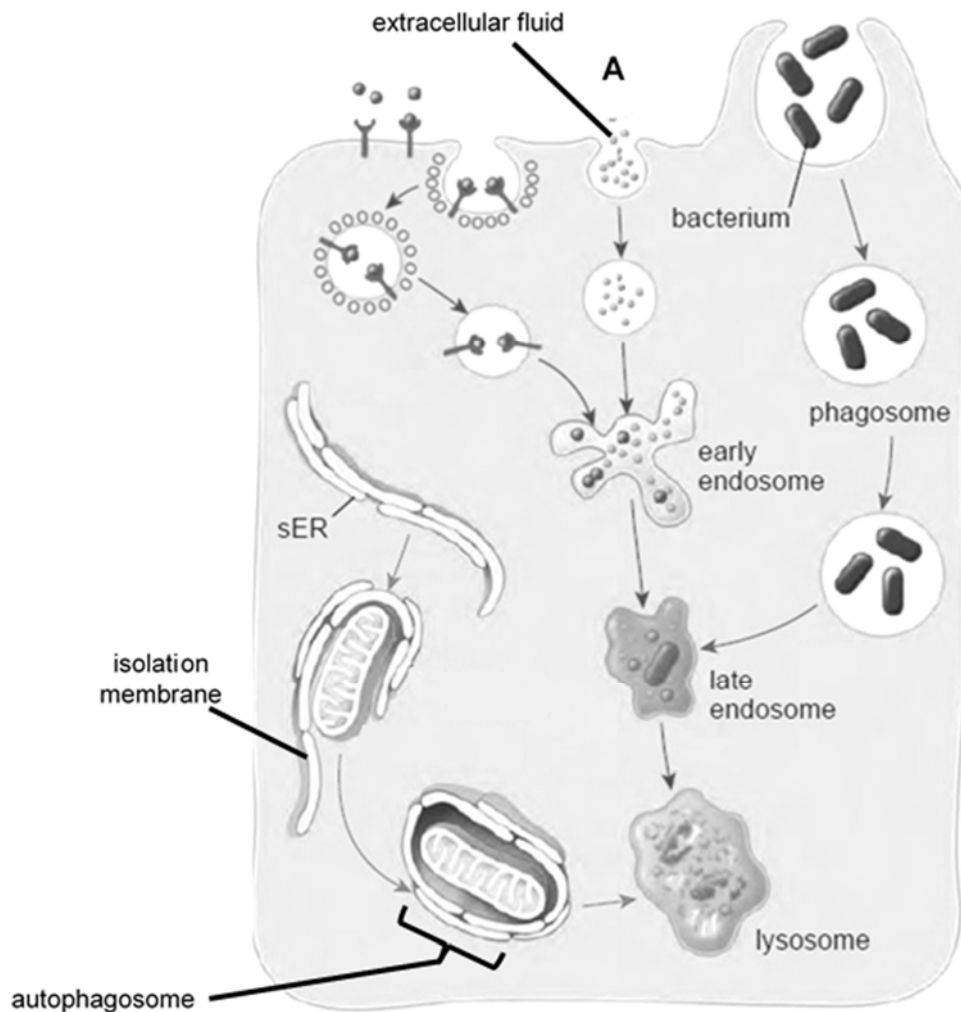


Fig. 1.1

- (a) (i) Identify precisely process **A**.

..... [1]

- (ii) State **one** property of the plasma membrane and explain how it enables process **A** to be carried out by a cell.

.....

 [2]

- (b) Degradation of worn out organelles such as mitochondria occurs inside most cells via autophagy. With reference to Fig.1.1, describe the process of autophagy.

.....

[3]

- (c) Lysosomes are able to hold a large amount of enzymes. Lysosomal membrane contains a large amount of highly glycosylated integral proteins facing the interior of the lysosome.

Suggest how this high amount of glycosylated protein prevents self-digestion of the lysosome.

..... [1]

Fig.1.2 shows an electron micrograph of parts of the endomembrane system.

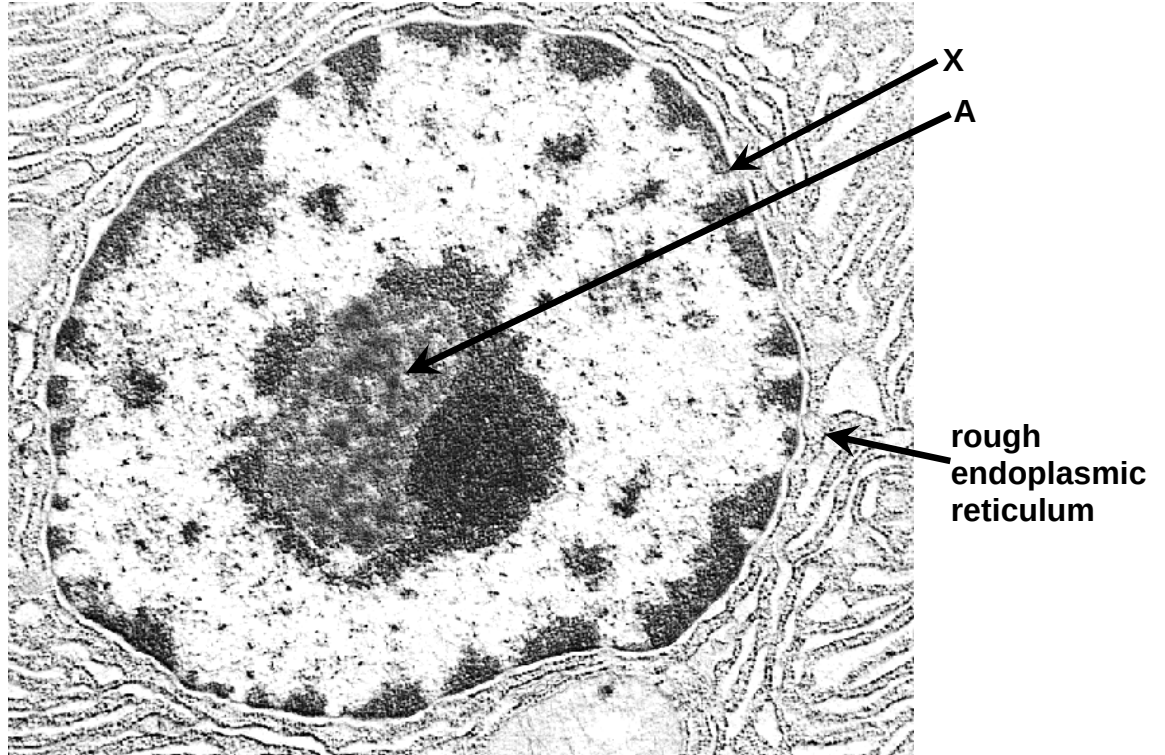


Fig. 1.2

- (d) Explain how **X** regulates the movement of materials in protein synthesis.

.....

 [2]

- (e) Explain how the functions of region **A** and rough endoplasmic reticulum are related.

.....

 [2]

- (f) During the early stages of oogenesis (formation of egg) in Xenopus laevis (frog); there are as many as 1000 of region **A** within a single oocyte (egg cell). Suggest the significance of this.

..... [1]

[Total: 12]

- 2 (a) In beer-making, barley is malted with enzymes which hydrolyse starch into sugar, ready for fermentation. The graph below shows the production of sugar during beer-making at three different temperatures over a period of 60 minutes. All other conditions were controlled.

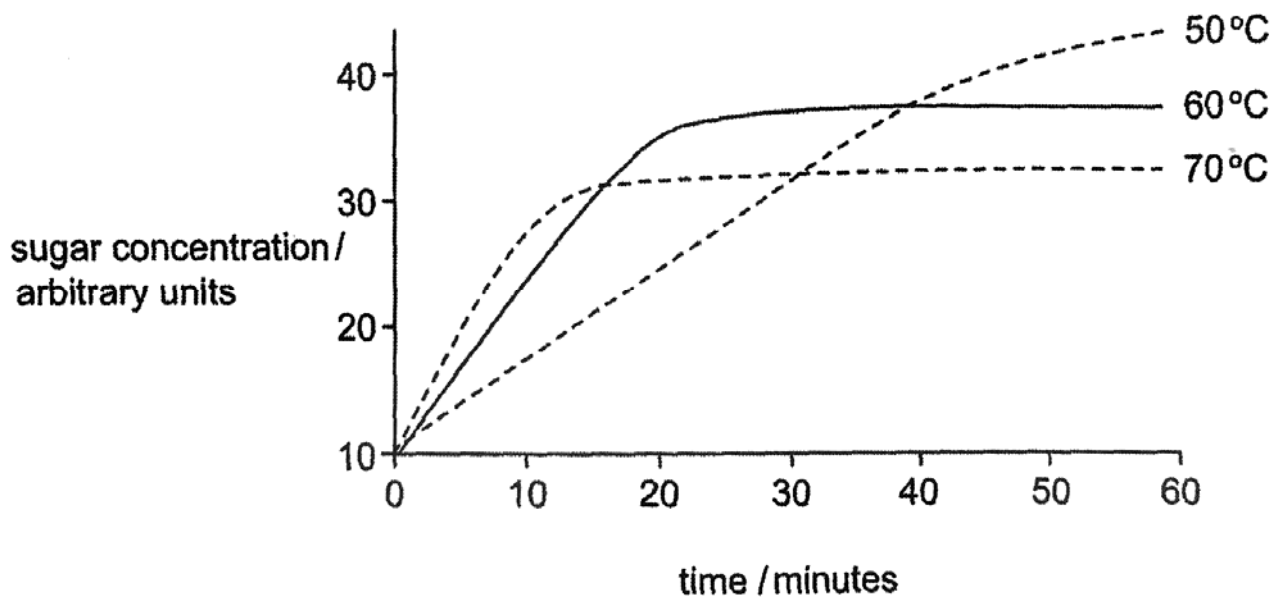


Fig. 2.1

- (i) With reference to Fig. 2.1, explain the effect of increasing temperature on enzyme activity for the first 10 minutes of the reaction.

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..... [3]

- (ii) Explain why the final concentration of sugar produced at 70°C is lower than the reaction incubated at 60°C.

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..... [2]

- (iii) State the enzyme used in the reaction above.

..... [1]

- (iv) Explain how the enzyme stated in (iii) plays its role in hydrolysis of starch to glucose.

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..... [3]

- (b) What structural differences exist between starch and cellulose, and how these are related to their different roles in plants.

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[Total: 12]

- 3 Fig. 3.1 below shows DNA replication in an eukaryotic organism.

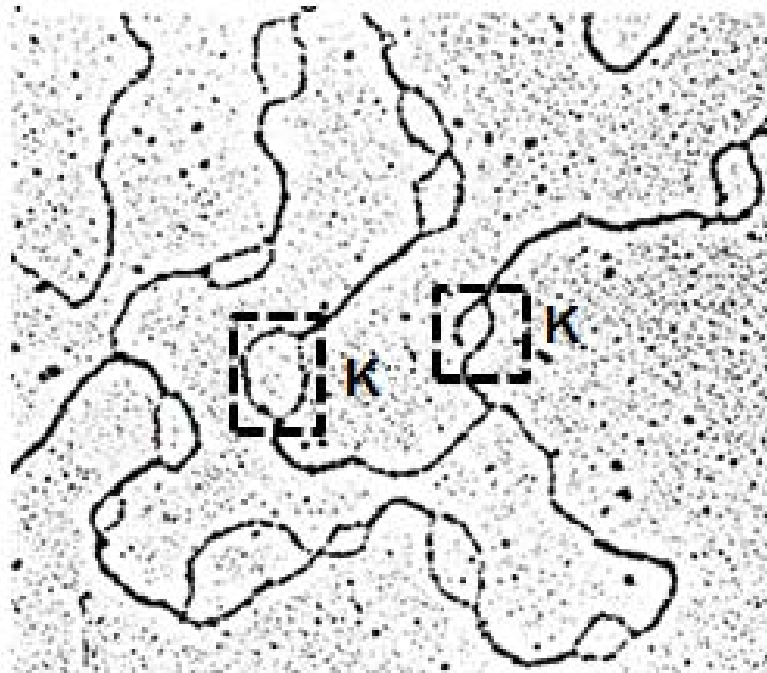


Fig. 3.1

- (a) (i) What evidence in Fig. 3.1 shows that the process is DNA replication in an eukaryotic cell.

.....
 [1]

- (ii) Within structures **K** in Fig. 3.1, there are no occurrences of end-replication problem. Explain why.

.....
 [1]

- (b) (i) The DNA replication at each replication fork is sometimes described as '*asymmetrical*' replication as there are differences in the way the daughter strands are being synthesized. State **two** such differences.

.....

.....

..... [2]

(ii) Suggest **two** reasons for the 'asymmetrical' replication.

.....

 [2]

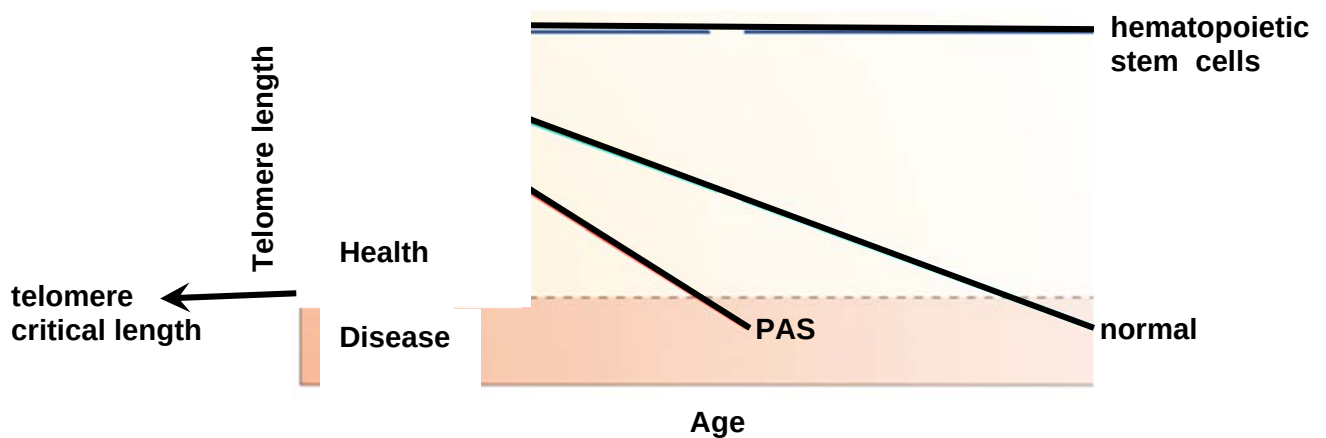
(c) Describe the unique features of hematopoietic stem cells which are common in all stem cells.

.....

 [3]

(d) Fig. 3.2 shows a graph showing the relationship between age and the telomere length in 3 different kinds of cell;

- hematopoietic stem cells
- somatic cells from a healthy individual (normal)
- somatic cells from an individual suffering premature aging syndromes (PAS)



Fig

With reference to

(i) account for the difference in telomere length between hematopoietic stem cells and somatic cells from healthy individual.

.....

-
- [2]
- (ii) suggest the cause of premature aging syndrome
-
-
-
- [2]

[Total: 13]

- 4 Root tissue from a barley seedling was prepared and its chromosomes were observed under a microscope. Fig. 4.1 shows a cell from the root tissue at the metaphase stage of mitosis.



Fig. 4.1

Fig. 4.2 shows the changes in amount of DNA at different stages of the barley life cycle.

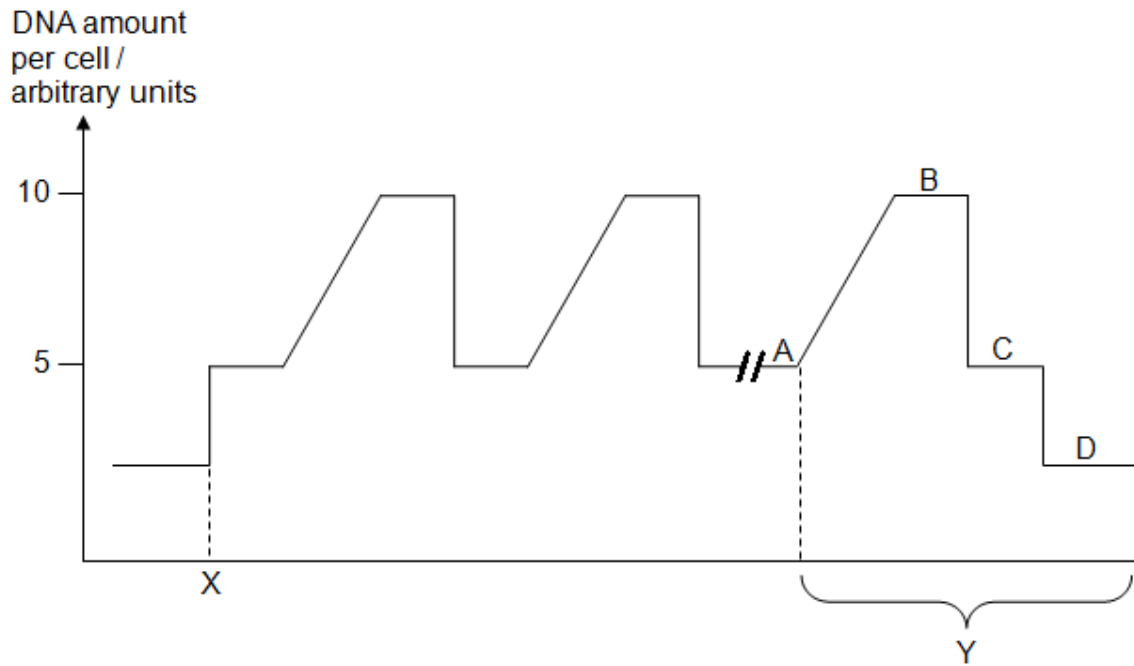


Fig. 4.2

- (a) Mark out clearly with an arrow  , on Fig. 4.2, the part of the graph which corresponds to the stage shown in Fig. 4.1.

[1]

- (b) With reference to Fig.4.2,

- (i) state which of the stages, from A to D has/ have the **same** number of chromosomes as shown in Fig.4.1.

..... [1]

- (ii) Explain why a mutation which occurs during Y is considered as a hereditary mutation.

..... [1]

- (c) Explain the significance of the event occurring at X.

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..... [2]

- (d) Fig. 4.3 shows the formation of Philadelphia chromosome which is commonly found in chronic myelogenous leukemia (CML) cells.

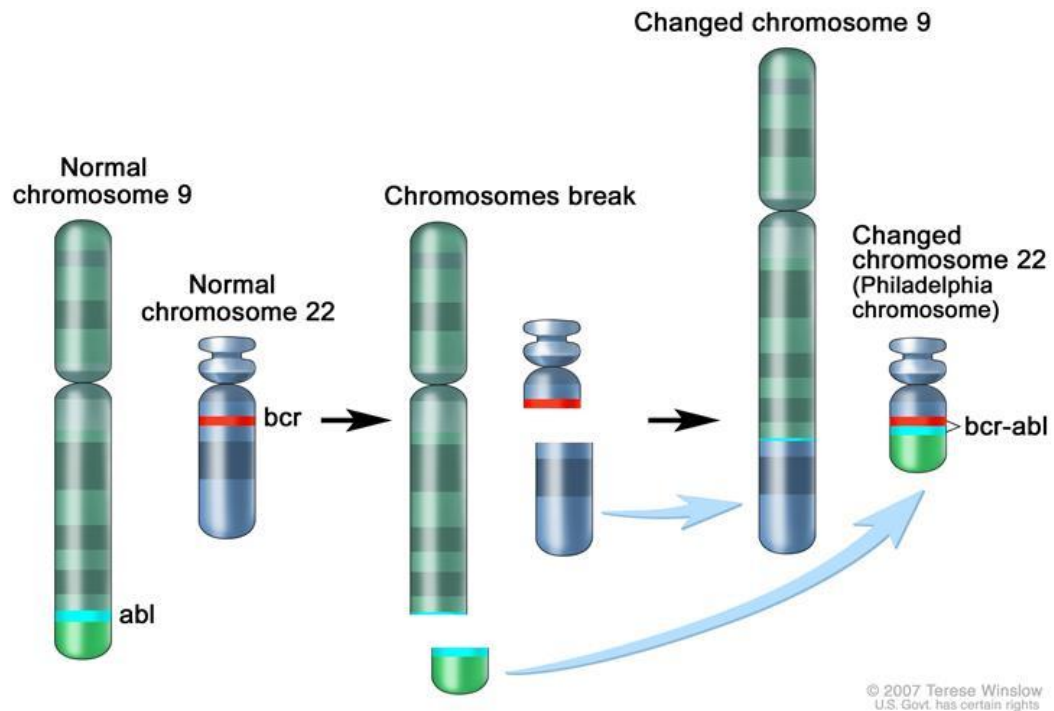


Fig. 4.3

- (i) With reference to Fig.4.3, describe the chromosome aberration which results in the formation of Philadelphia chromosome.

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..... [2]

The protein product of *bcr* gene (BCR protein) is a protein involved in signaling pathway and possesses tyrosine kinase activity. In the presence of growth factors, BCR protein is activated and is found to promote cell growth and proliferation.

The results of a study conducted using chronic myelogenous leukemia (CML) cells to show the effect of Philadelphia chromosome on the activity of BCR protein is shown in Table 4.1.

Table 4.1

The variable being studied	normal cells	CML cells
Concentration of BCR protein / mg cm^{-3}	29.8	29.5
Tyrosine kinase activity of BCR protein in the absence of growth factor / a.u.	0	40
Tyrosine kinase activity of BCR protein in the presence of growth factor/ a.u.	40	40

- (ii) With respect to its effect on cell growth and proliferation, name the group of genes which *bcr* gene belongs to.

..... [1]

- (iii) With reference to Fig. 4.3 and Table 4.1, explain how Philadelphia chromosome contributes to the onset of chronic myelogenous leukemia (CML).

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.....[4]

[Total: 12]

- 5 In a particular variety of tomato plant, the allele for red fruit colour is dominant to the allele for yellow fruit colour and the allele for hairy stems is codominant with the allele for hairless stems. A true breeding plant with red fruit and hairy stems was crossed with another true breeding plant with yellow fruit and hairless stems. The resulting F1 were selfed to produce one hundred tomato plants with their ratios shown in Table 5.1.

Table 5.1

Frequency and phenotype of offspring	Genotype
37 red fruit and short hairs on stem	
18 red fruit and very hairy stem	
19 red fruit and hairless stem	
13 yellow fruit and short hairs on stem	
7 yellow fruit and very hairy stem	
6 yellow fruit and hairless stem	

- (a) Using the letters **R** for red fruit and **r** for yellow fruit, **H** for hairy stem and **L** for hairless stem, fill in the genotypes for each phenotype of the offspring in the Table 5.1 above. [2]
- (b) From Table 5.1, explain how codominance brings about the trait “short hairs on stem”.

[2]
- (c) Draw a genetic diagram to explain this cross.

[Total 9]

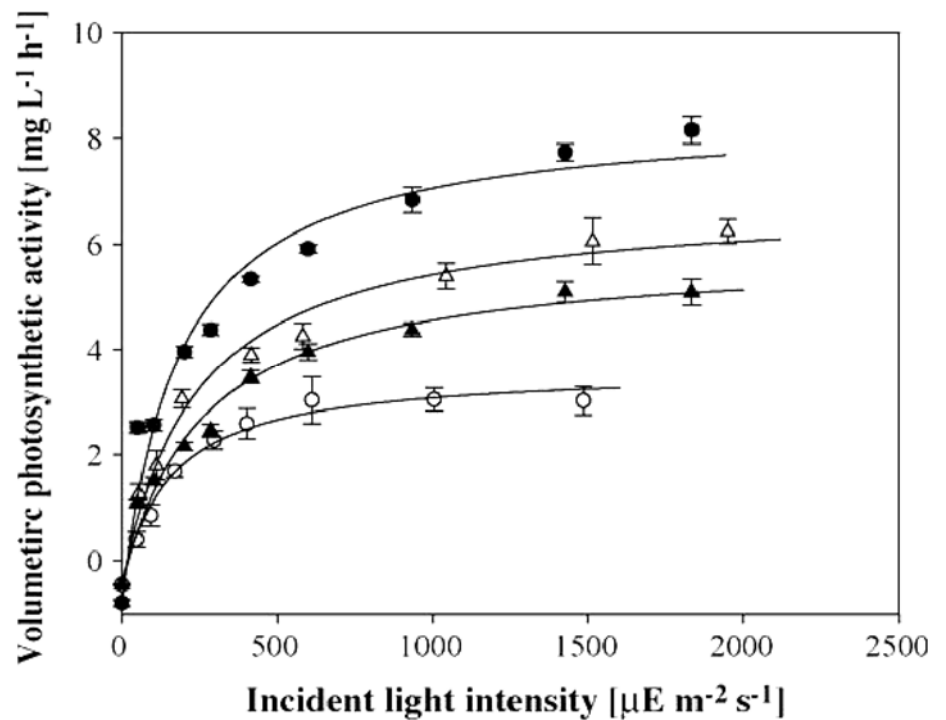
- 6 Microalgae have been extensively studied for various purposes, such as the production of biomass as a source of valuable chemicals of health foods and for wastewater treatment. Recently, microalgal photosynthesis was considered to be an effective means to reduce the emission of carbon dioxide, a major greenhouse gas, in the atmosphere. Light is the most important factor affecting microalgal photosynthesis kinetics. In general, most microalgal mass culture systems are limited by light, because light is easily absorbed and scattered by the microalgal cells. Therefore, understanding and quantification of light dependence of microalgal activity is of great importance in designing an efficient photobioreactor, in predicting process performance, and in optimizing operating conditions.

Fig. 6.1

The volumetric photosynthetic activity as a function of incident light intensity at different light types and cell concentrations. Data points and error bars were average values and standard deviations of three replicated experimental results. Solid lines represent the calculated results from the photosynthesis–irradiance model. The light types and cell concentrations were:

(○) simulated daylight and 0.215 g L⁻¹;
 (▲) simulated daylight and 0.123 g L⁻¹;
 (△) red light and 0.123 g L⁻¹; and
 (+) green light and 0.123 g L⁻¹.

Jeon *et al.* 2005 Measurement of microalgal photosynthetic activity depending on light intensity and quality. *Biochemical Engineering Journal* 27 (2005) 127–131



- (a) Explain the trends seen when red, green and daylight (at 0.123g/L) are compared.

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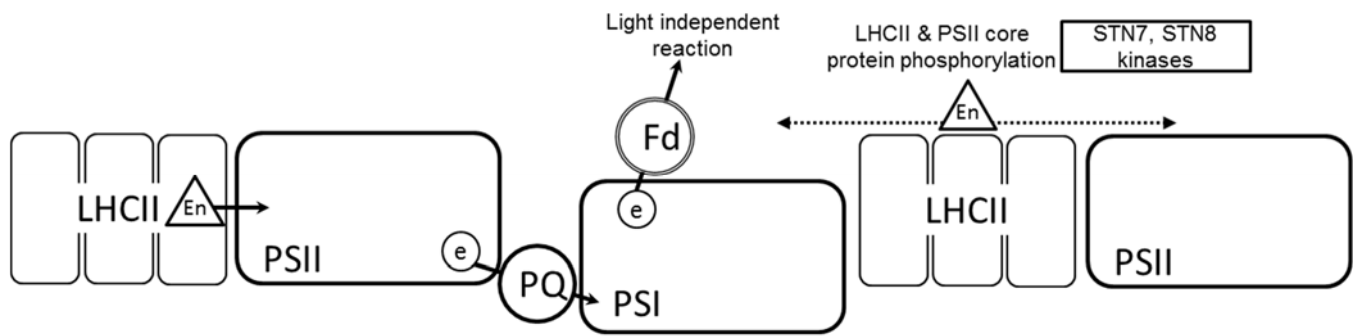
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.....[5]

Fig. 6.2 shows a schematic showing the functional relationship between light harvesting complexes (LHC) and photosystems II & I. Regulatory complexes are also shown comprising of kinases and the regulation of excess energy between PS II and I.



Gollan et al 2015 Photosynthetic light reactions: integral to chloroplast retrograde signalling. Current Opinion in Plant Biology 27:180-191modified.

Fig. 6.2

(b) Explain what is the LHC and its role in photosynthesis.

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.....[2]

(c) With reference to Fig. 6.2 explain the role of electrons in the photosynthesis as they move from Photosystem II to Photosystem I.

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.....[3]

(d) With reference to Fig. 6.2 suggest the implications of the role of LHC and PSII core protein phosphorylation from Photosystem II to Photosystem I.

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.....[3]

[Total: 13]

- 7 The Isthmus of Panama is the narrow strip of land that lies between the Caribbean Sea and the Pacific Ocean, linking North and South America. It contains the country of Panama and the Panama Canal. The isthmus was formed around 2.8 million years ago. This major geological event separated the Atlantic and Pacific Oceans and caused the creation of the Gulf Stream.

The genus Anisotremus shown in Fig. 7.1b comprises 9 described species which occur predominantly on coral reefs and subtropical rocky reefs in the Neotropics of the Tropical Eastern Pacific the Caribbean and adjacent waters. In this study, the phylogenetic relationships for all described species were examined based on one mitochondrial gene (cytochrome b) and one nuclear marker (the first intron of the ribosomal protein S7).

Fig 7.1a

Molecular ecology, speciation, and evolution of the reef fish genus Anisotremus
Bernardi et al *Molecular Phylogenetics and Evolution* 48 (2008) 929–935

Fig 7.1a

- (a) Name two methods by which evolution can take place.

.....

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.....[2]

Fig 7.1b

- (b) With reference to Fig. 7.1b explain the type of speciation that would have been seen to the derivation of the two fish A. virginicus and A. taeniatus.

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.....[2]

- (c) With reference to your answer in (b) explain how micro evolution would have taken place.

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.....[4]

- (d) Suggest how it was determined that A. virginicus and A. taeniatus were phylogenetically descended from A. dovii and A. pacifici.

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.....[2]

- (e) In this study it was proposed that A. virginicus and A. taeniatus took a shorter time to speciate from one another compared to A. dovii and A. pacifici. Suggest with evidence from Fig. 7.1a how this might be true.

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.....[2]

- 8 Transpeptidase is a bacterial enzyme that cross-links cell wall peptides during the formation of bacterial cell walls. The antibiotic penicillin inhibits the activity of transpeptidase. Fig. 8.1 shows part of each of the molecular structures of a cell wall peptide and penicillin.

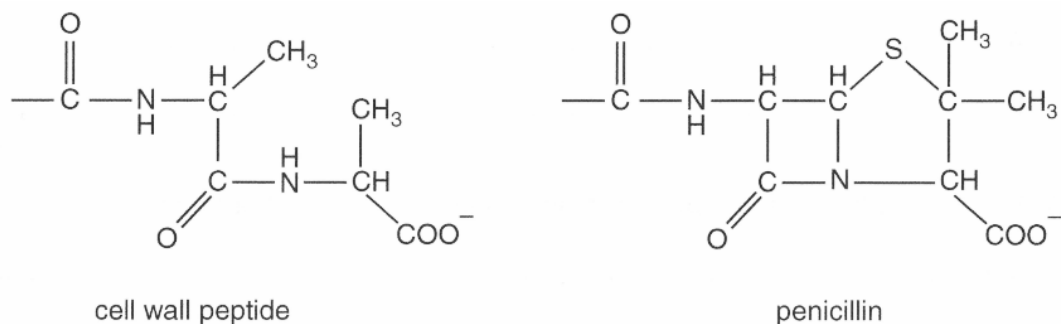


Fig. 8.1

- (a) Comment on the structure of cell wall peptides and penicillin.

.....[1]

- (b) Suggest why the penicillin molecule is an effective inhibitor of transpeptidase.

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[2]

- (c) Fig.8.2 shows an electron micrograph of an alveolar macrophage isolated from a tuberculosis patient.

M. tuberculosis

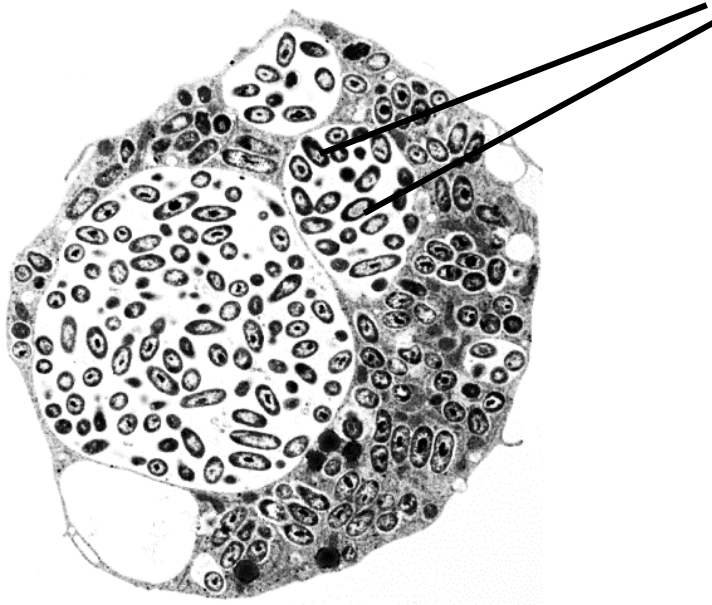


Fig. 8.2

- (i) Describe the mode of transmission of Mycobacterium tuberculosis.

..... [1]

- (ii) Explain the appearance of the alveolar macrophage in Fig. 8.2.

..... [2]

- (d) Tuberculosis patients are commonly treated with antibiotics, isoniazid and rifampicin. Recently, there is an increase in number of multi-drug resistant tuberculosis cases. State **one** reason why multi-drug resistant tuberculosis continues to emerge.

..... [1]
[Total 7]

- 9 Dengue is the most rapidly spreading mosquito-borne viral disease in the world. In the last 50 years, incidence has increased 30-fold with increasing geographic expansion to new countries and, in the present decade, from urban to rural settings (Fig. 9.1). An estimated 50 million dengue infections occur annually and approximately 2.5 billion people live in dengue endemic countries.



Fig. 9.1 Shaded areas are countries at risk of dengue fever due to presence of Aedes mosquito, as of 2008. The contour lines are range of January/July isotherm indicating the potential range of Aedes aegypti.

(a) Describe the developmental stages (including duration) in the life cycle of the Aedes mosquito.

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.....[4]

(b) Explain why the range of dengue fever is the same as that of the Aedes mosquito.

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.....[2]

(c) To some extent the range of the Aedes mosquito has also followed human expansion, explain how this may be true.

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.....[2]

(d) With reference to Fig. 9.1, explain how climate change may affect the spread of dengue beyond the tropics.

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.....	
.....	[2]
	[Total: 10]

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